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Set 14: Survey Results Analysis

Data Visualization using R

Aim

To analyse survey results using a stacked bar chart, a radar chart, a table and a dashboard in R.

Procedure

1. Open RStudio and load the packages ggplot2 and gridExtra (install them once with `install.packages(c("ggplot2", "gridExtra"))`).
2. Create the survey data as a data frame.
3. Question 1: draw the stacked bar chart of Question 1 with `geom_bar()`.
4. Question 2: convert answers to numbers (A = 1 to D = 4) and draw the radar chart with `coord_polar()`.
5. Question 3: make the table with `tableGrob()`.
6. Question 4: join the charts and the table into a dashboard with `arrangeGrob()`.
7. Run the code and note the output and the figures.

R Code

```
library(ggplot2); library(gridExtra)

# Dataset
survey <- data.frame(Respondent = 1:3, Question1 = c("A","B","C"),
                     Question2 = c("B","A","A"), Question3 = c("C","D","B"))
print(survey)

# 1. Stacked bar chart - Question 1
p1 <- ggplot(survey, aes("Question 1", fill = Question1)) + geom_bar() +
  labs(title = "Answers of Question 1", x = "", y = "Count", fill = "Answer")

# 2. Radar chart (A=1, B=2, C=3, D=4)
score <- sapply(survey[, 2:4], match, LETTERS)
score <- cbind(score, score[, 1]) # repeat Q1 to close the shape
radar <- data.frame(Respondent = factor(rep(1:3, 4)), Q = rep(1:4, each = 3),
                   Score = as.vector(score))
p2 <- ggplot(radar, aes(Q, Score, group = Respondent, colour = Respondent)) +
  geom_polygon(fill = NA) + geom_point(size = 2) + coord_polar() +
  scale_x_continuous(limits = c(1, 4), breaks = 1:3, labels = c("Q1","Q2","Q3")) +
  scale_y_continuous(limits = c(0, 4), breaks = 1:4, labels = c("A","B","C","D")) +
  labs(title = "Radar Chart of Responses", x = "", y = "")

# 3. Table
tb <- tableGrob(survey, rows = NULL)

# 4. Dashboard
dash <- arrangeGrob(p1, p2, tb, layout_matrix = rbind(c(1,2), c(3,3)), top = "Survey Results Dashboard")

ggsave("set14_q1.png", p1, width = 6, height = 4, bg = "white")
ggsave("set14_q2.png", p2, width = 6, height = 5, bg = "white")
ggsave("set14_q3.png", arrangeGrob(tb, top = "Survey Results Table"), width = 5, height = 2, bg = "white")
```

```
ggsave("set14_q4.png", dash, width = 12, height = 8, bg = "white")
```

Output

	Respondent	Question1	Question2	Question3
1	1	A	B	C
2	2	B	A	D
3	3	C	A	B

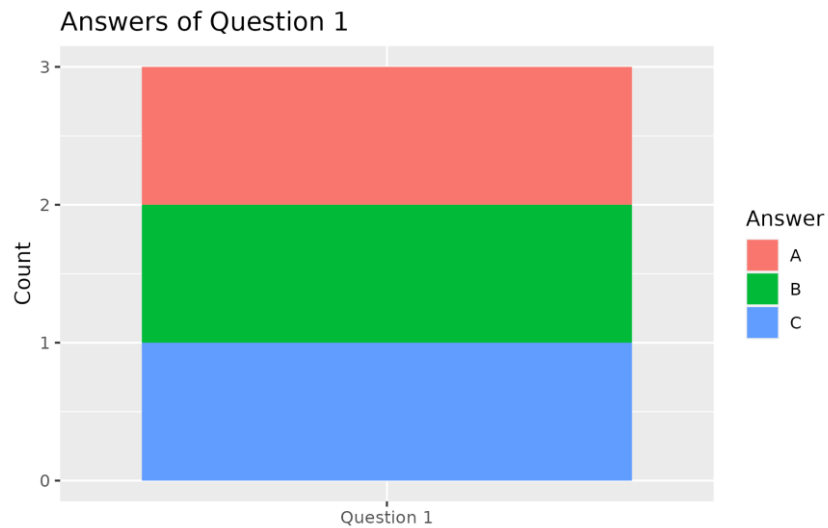


Figure 14.1: Q1 - Stacked bar chart (Question 1)

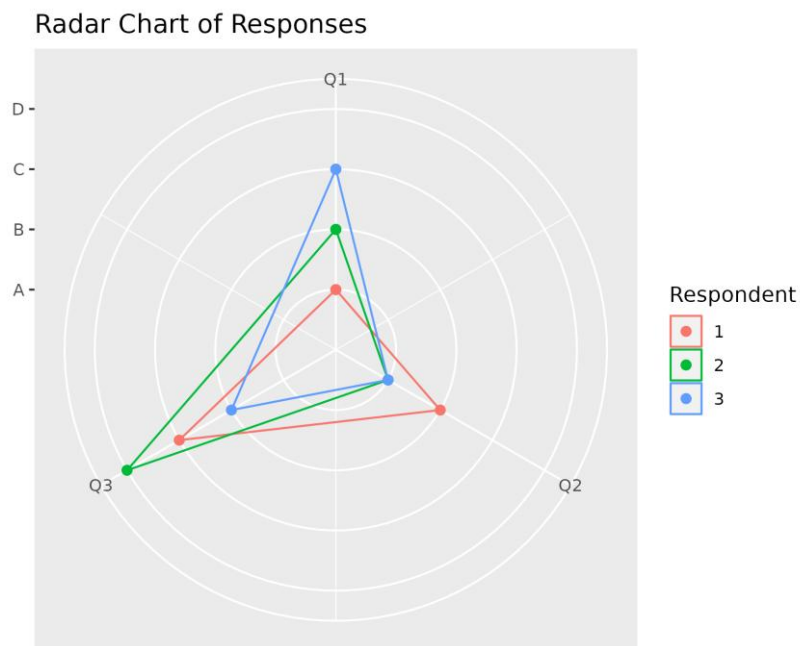
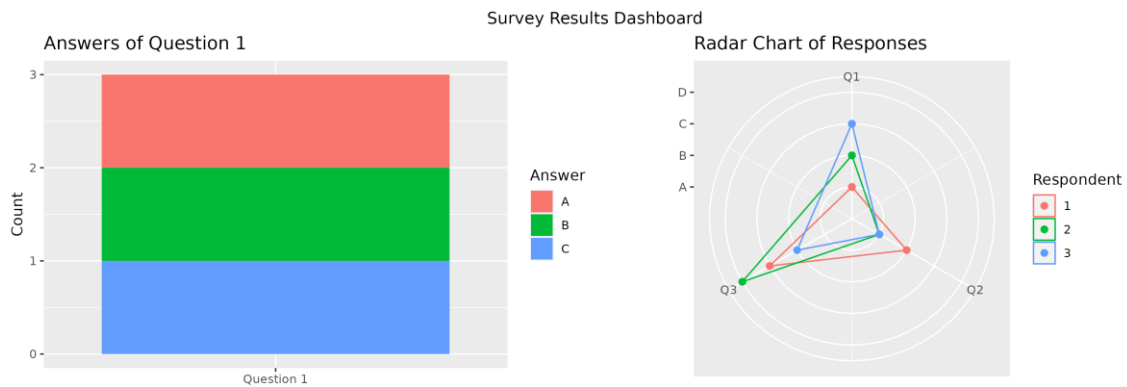


Figure 14.2: Q2 - Radar chart (A=1, B=2, C=3, D=4)

Survey Results Table

Respondent	Question1	Question2	Question3
1	A	B	C
2	B	A	D
3	C	A	B

Figure 14.3: Q3 - Survey table



Respondent	Question1	Question2	Question3
1	A	B	C
2	B	A	D
3	C	A	B

Figure 14.4: Q4 - Dashboard

Result

- Each answer (A, B, C) in Question 1 was chosen once.
- The radar chart shows that the three respondents have different response patterns.
- All four parts were completed successfully in R.

Set 15: Product Sales Analysis

Data Visualization using R

Aim

To analyse monthly product sales using a grouped bar chart, a stacked area chart, a table and a dashboard in R.

Procedure

8. Open RStudio and load the packages ggplot2 and gridExtra (install them once with `install.packages(c("ggplot2", "gridExtra"))`).
9. Create the sales data as a data frame and convert it to long form.
10. Question 1: draw the grouped bar chart with `geom_col(position = "dodge")`.
11. Question 2: draw the stacked area chart with `geom_area()`.
12. Question 3: make the table with `tableGrob()`.
13. Question 4: join the charts and the table into a dashboard with `arrangeGrob()`.
14. Run the code and note the output and the figures.

R Code

```
library(ggplot2); library(gridExtra)

# Dataset
sales <- data.frame(Product_ID = 1:3, Product = c("Product A", "Product B", "Product C"),
  Jan_Sales = c(2000, 1500, 1200), Feb_Sales = c(2200, 1800, 1400),
  Mar_Sales = c(2400, 1600, 1100))
print(sales)
long <- data.frame(Product = rep(sales$Product, 3),
  Month = rep(1:3, each = 3),
  Sales = c(sales$Jan_Sales, sales$Feb_Sales, sales$Mar_Sales))
print(aggregate(Sales ~ Product, long, sum))
print(aggregate(Sales ~ Month, long, sum))

# 1. Grouped bar chart
p1 <- ggplot(long, aes(Product, Sales, fill = factor(Month, labels = c("Jan", "Feb", "Mar")))) +
  geom_col(position = "dodge") + labs(title = "Q1 Sales of Each Product", y = "Sales", fill = "Month")

# 2. Stacked area chart
p2 <- ggplot(long, aes(Month, Sales, fill = Product)) + geom_area() +
  scale_x_continuous(breaks = 1:3, labels = c("Jan", "Feb", "Mar")) +
  labs(title = "Overall Sales Trend", x = "Month", y = "Total Sales")

# 3. Table
tb <- tableGrob(sales, rows = NULL)

# 4. Dashboard
dash <- arrangeGrob(p1, p2, tb, layout_matrix = rbind(c(1,2), c(3,3)), top = "Product Sales Dashboard")

ggsave("set15_q1.png", p1, width = 6, height = 4, bg = "white")
ggsave("set15_q2.png", p2, width = 6, height = 4, bg = "white")
ggsave("set15_q3.png", arrangeGrob(tb, top = "Monthly Sales Table"), width = 6, height = 2, bg = "white")
ggsave("set15_q4.png", dash, width = 12, height = 8, bg = "white")
```

Output

	Product_ID	Product	Jan_Sales	Feb_Sales	Mar_Sales
1	1	Product A	2000	2200	2400
2	2	Product B	1500	1800	1600
3	3	Product C	1200	1400	1100

Product Sales

```

1 Product A 6600
2 Product B 4900
3 Product C 3700
Month Sales
1 1 4700
2 2 5400
3 3 5100

```

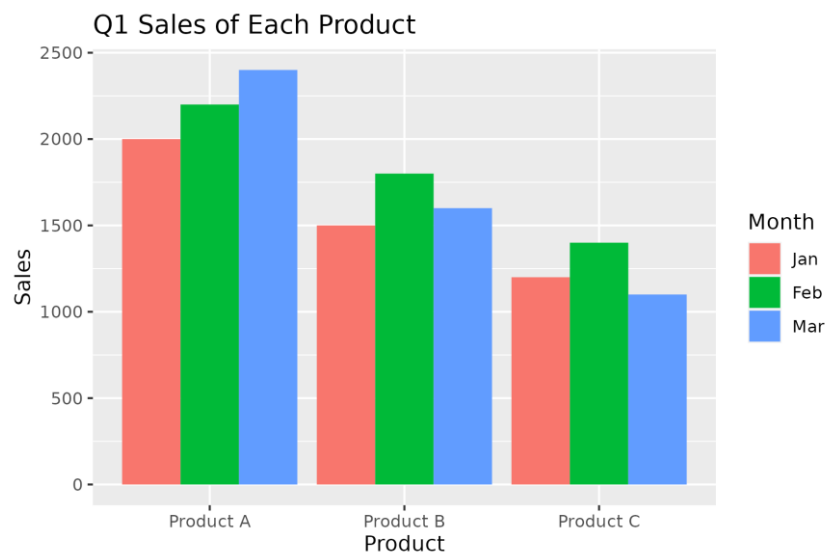


Figure 15.1: Q1 - Grouped bar chart

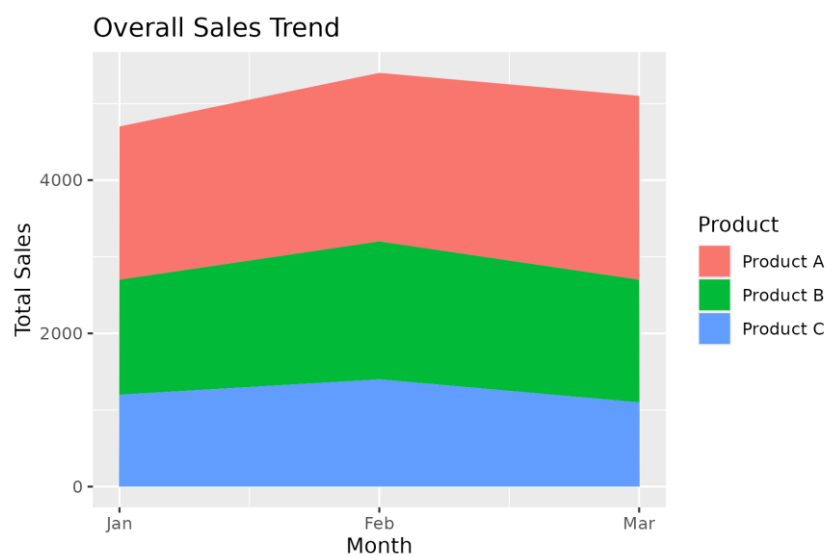
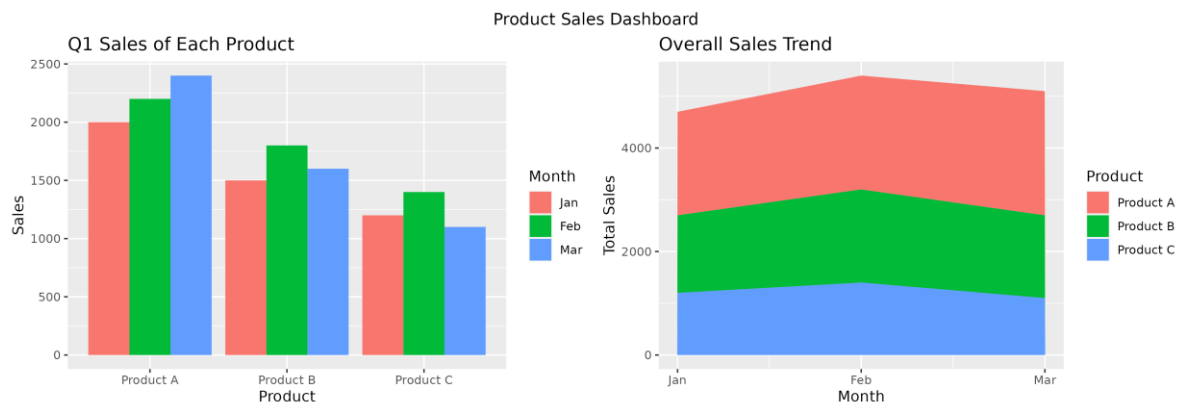


Figure 15.2: Q2 - Stacked area chart

Monthly Sales Table

Product_ID	Product	Jan_Sales	Feb_Sales	Mar_Sales
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

Figure 15.3: Q3 - Monthly sales table



Product_ID	Product	Jan_Sales	Feb_Sales	Mar_Sales
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

Figure 15.4: Q4 - Dashboard

Result

- Quarter totals: Product A = 6600, Product B = 4900, Product C = 3700.
- Total sales were highest in February (5400).
- All four parts were completed successfully in R.

Set 16: Customer Demographics Analysis

Data Visualization using R

Aim

To analyse customer demographics using a bar chart, a pie chart, a table and a dashboard in R.

Procedure

15. Open RStudio and load the packages ggplot2 and gridExtra (install them once with `install.packages(c("ggplot2", "gridExtra"))`).
16. Create the customer data as a data frame and group ages with `cut()`.
17. Question 1: draw the age bar chart with `geom_bar()`.
18. Question 2: draw the gender pie chart with `geom_col()` and `coord_polar("y")`.
19. Question 3: make the table with `tableGrob()`.
20. Question 4: join the charts and the table into a dashboard with `arrangeGrob()`.
21. Run the code and note the output and the figures.

R Code

```
library(ggplot2); library(gridExtra)

# Dataset
cust <- data.frame(Customer_ID = 1:3, Age = c(28, 35, 42),
  Gender = c("Female", "Male", "Female"), Income = c(50000, 60000, 75000))
print(cust)
cust$Age_Group <- cut(cust$Age, c(20, 30, 40, 50), right = FALSE,
  labels = c("20-29", "30-39", "40-49"))
gender <- as.data.frame(table(cust$Gender)); names(gender) <- c("Gender", "Count")
print(gender)

# 1. Bar chart - age distribution
p1 <- ggplot(cust, aes(Age_Group)) + geom_bar(fill = "steelblue") +
  labs(title = "Customer Age Distribution", x = "Age Group", y = "Customers")

# 2. Pie chart - gender
p2 <- ggplot(gender, aes("", Count, fill = Gender)) + geom_col() + coord_polar("y") +
  geom_text(aes(label = Count), position = position_stack(vjust = 0.5)) +
  labs(title = "Customers by Gender") + theme_void()

# 3. Table
tb <- tableGrob(cust[, 1:4], rows = NULL)

# 4. Dashboard
dash <- arrangeGrob(p1, p2, tb, layout_matrix = rbind(c(1,2), c(3,3)), top = "Customer Dashboard")

ggsave("set16_q1.png", p1, width = 6, height = 4, bg = "white")
ggsave("set16_q2.png", p2, width = 6, height = 4, bg = "white")
ggsave("set16_q3.png", arrangeGrob(tb, top = "Customer Demographics Table"), width = 5, height = 2, bg = "white")
ggsave("set16_q4.png", dash, width = 12, height = 8, bg = "white")
```

Output

```
Customer_ID Age Gender Income
1      1  28 Female  50000
2      2  35  Male  60000
3      3  42 Female  75000
Gender Count
1 Female    2
2 Male      1
```

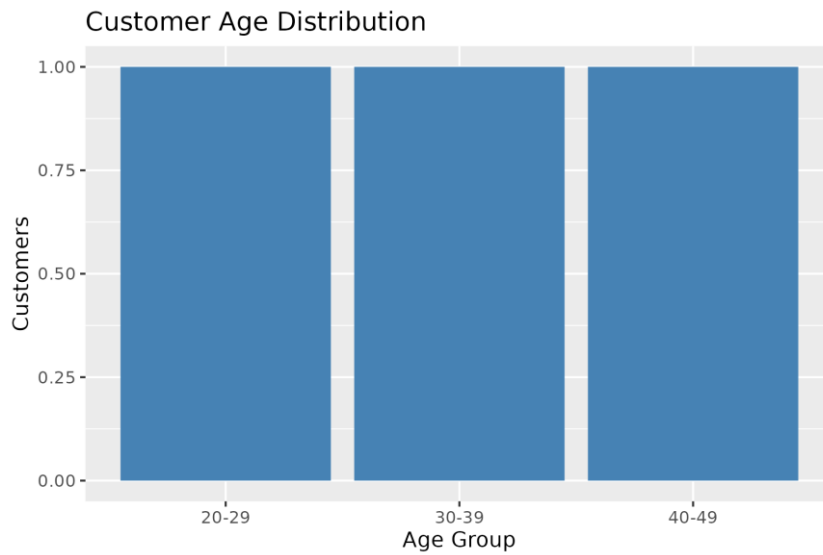


Figure 16.1: Q1 - Age distribution bar chart

Customers by Gender

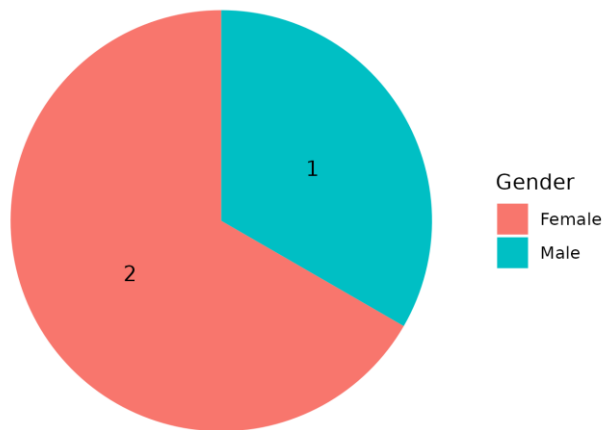
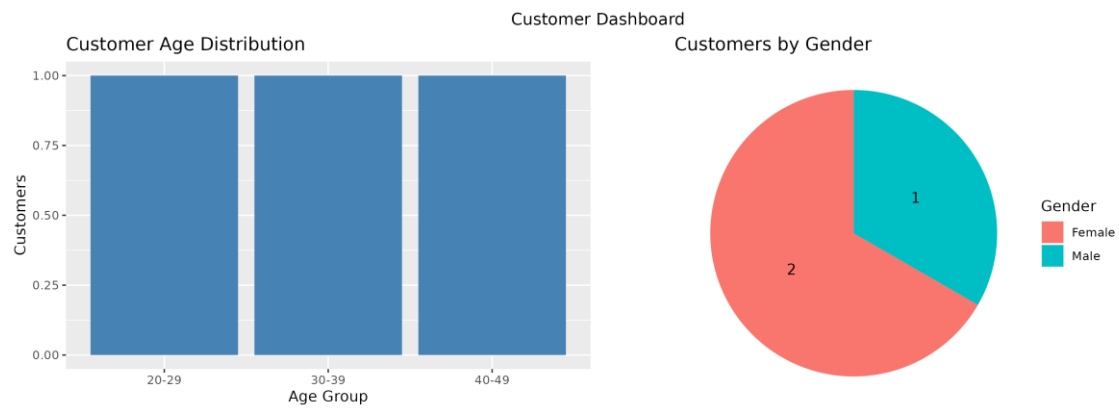


Figure 16.2: Q2 - Gender pie chart

Customer Demographics Table

Customer_ID	Age	Gender	Income
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

Figure 16.3: Q3 - Customer table



Customer_ID	Age	Gender	Income
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

Figure 16.4: Q4 - Dashboard

Result

- There is one customer in each age group (20-29, 30-39, 40-49).
- Two customers are female and one is male.
- All four parts were completed successfully in R.

Set 17: Employee Performance Analysis

Data Visualization using R

Aim

To analyse employee performance using a line chart, a bar chart, a table and a dashboard in R.

Procedure

22. Open RStudio and load the packages ggplot2 and gridExtra (install them once with `install.packages(c("ggplot2", "gridExtra"))`).
23. Create the employee data as a data frame.
24. Question 1: draw the line chart of performance score against years of service with `geom_line()`.
25. Question 2: draw the bar chart of employees per department with `geom_bar()`.
26. Question 3: make the table with `tableGrob()`.
27. Question 4: join the charts and the table into a dashboard with `arrangeGrob()`.
28. Run the code and note the output and the figures.

R Code

```
library(ggplot2); library(gridExtra)

# Dataset
emp <- data.frame(Employee_ID = 1:3, Department = c("Sales", "HR", "Marketing"),
                  Years = c(5, 3, 7), Score = c(85, 92, 78))
print(emp)
cat("Average score:", mean(emp$Score), "\n")

# 1. Line chart - performance over years of service
p1 <- ggplot(emp, aes(Years, Score)) + geom_line() + geom_point(size = 3) +
  geom_text(aes(label = Department), vjust = -1) +
  labs(title = "Performance vs Years of Service", x = "Years of Service", y = "Performance Score")

# 2. Bar chart - employees per department
p2 <- ggplot(emp, aes(Department, fill = Department)) + geom_bar(show.legend = FALSE) +
  labs(title = "Employees per Department", x = "Department", y = "Employees")

# 3. Table
tb <- tableGrob(emp, rows = NULL)

# 4. Dashboard
dash <- arrangeGrob(p1, p2, tb, layout_matrix = rbind(c(1,2), c(3,3)), top = "Employee Performance Dashboard")

ggsave("set17_q1.png", p1, width = 6, height = 4, bg = "white")
ggsave("set17_q2.png", p2, width = 6, height = 4, bg = "white")
ggsave("set17_q3.png", arrangeGrob(tb, top = "Employee Performance Table"), width = 5, height = 2, bg = "white")
ggsave("set17_q4.png", dash, width = 12, height = 8, bg = "white")
```

Output

```
Employee_ID Department Years Score
1         1     Sales      5     85
2         2       HR       3     92
3         3 Marketing      7     78
Average score: 85
```

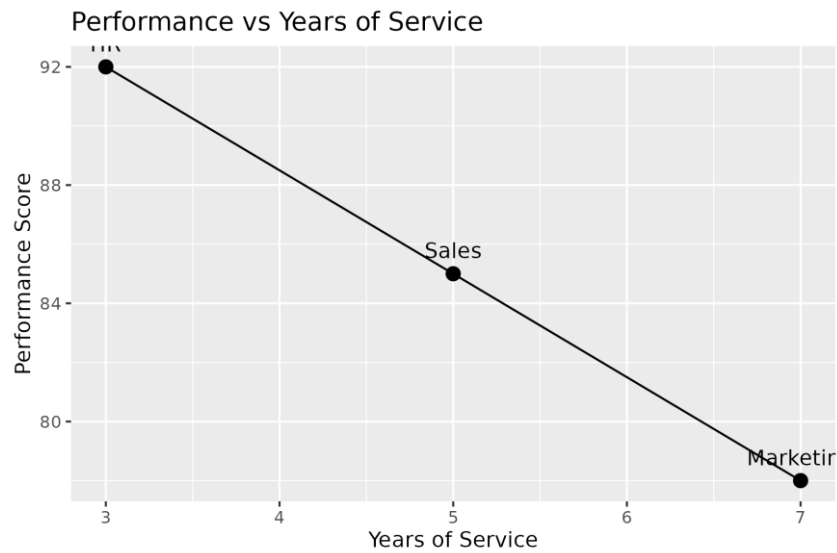


Figure 17.1: Q1 - Performance line chart

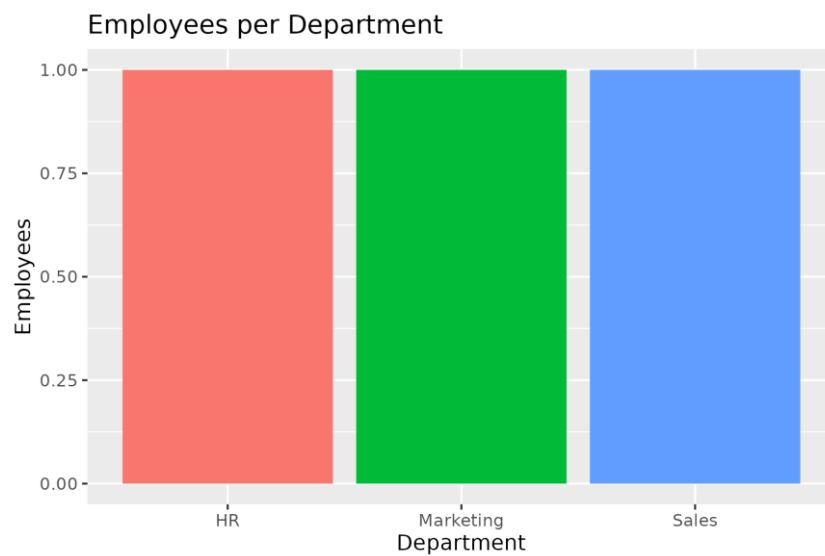
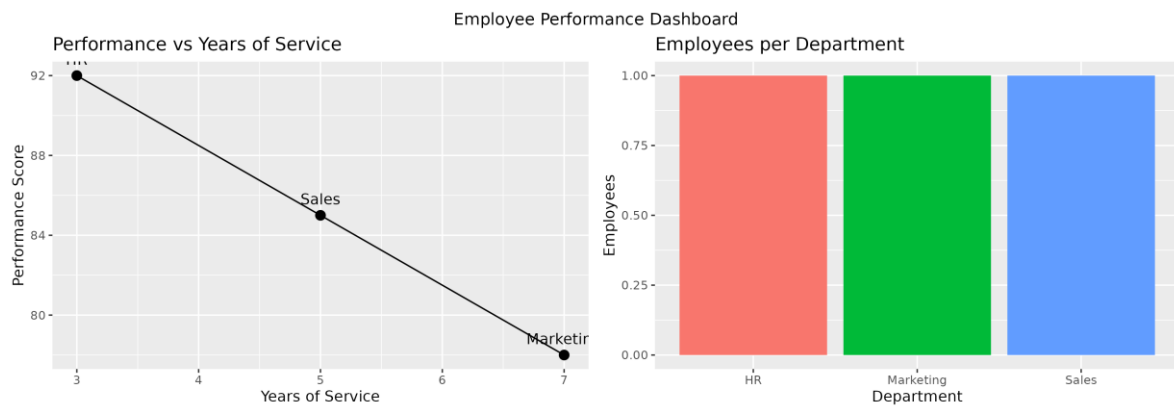


Figure 17.2: Q2 - Employees per department

Employee Performance Table

Employee_ID	Department	Years	Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

Figure 17.3: Q3 - Employee performance table



Employee_ID	Department	Years	Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

Figure 17.4: Q4 - Dashboard

Result

- The average performance score is 85.
- The score falls as years of service increase (92, 85, 78), but there are only 3 employees.
- Each department has one employee.